

```

Val Coverage: (N_b=261, N_r=236, N_r/N_b=0.9042)
Train Coverage: (N_b=4515, N_r=4119, N_r/N_b=0.9123)

[8] intergovorgs3(X1, X2) :- relintergovorgs(X1, X2) and intergovorgs(X2, X1) and
    timesinceally(X2, X1) and not(exporthbooks(X1, X2)) and not(dependent(X3, X1)) and not(
    warning(X1, X3)).
Training Acc: 0.6166 | Eval Acc: 0.6130
Val Coverage: (N_b=256, N_r=231, N_r/N_b=0.9023)
Train Coverage: (N_b=4676, N_r=4286, N_r/N_b=0.9166)

```

NEGATIVECOMM(X_1, X_2)

The target predicate `negativecomm( $X_1, X_2$ )` models hostile or adversarial communication between entities in the Nations knowledge base. Below, we present representative rules extracted by ANDRE, after removing redundant symmetric variants and rules with overlapping recursive structure.

Listing 5: Extracted Rules for `negativecomm( $X_1, X_2$ )` (Nations Dataset)

```

[1] negativecomm(X1, X2) :- negativebehavior(X1, X2) and timesinceally(X2, X1).
Training Acc: 0.9084 | Eval Acc: 0.9223
Val Coverage: (N_b=124, N_r=124, N_r/N_b=1.0000)
Train Coverage: (N_b=2228, N_r=2228, N_r/N_b=1.0000)

[2] negativecomm(X1, X2) :- negativebehavior(X1, X2) and accusation(X1, X2).
Training Acc: 0.9208 | Eval Acc: 0.9142
Val Coverage: (N_b=141, N_r=129, N_r/N_b=0.9149)
Train Coverage: (N_b=2799, N_r=2615, N_r/N_b=0.9343)

[3] negativecomm(X1, X2) :- negativebehavior(X1, X2) and blockpositionindex(X1, X2).
Training Acc: 0.8974 | Eval Acc: 0.9049
Val Coverage: (N_b=137, N_r=123, N_r/N_b=0.8978)
Train Coverage: (N_b=2411, N_r=2229, N_r/N_b=0.9245)

[4] negativecomm(X1, X2) :- negativebehavior(X1, X2) and pprotests(X1, X2).
Training Acc: 0.8855 | Eval Acc: 0.9026
Val Coverage: (N_b=107, N_r=107, N_r/N_b=1.0000)
Train Coverage: (N_b=1853, N_r=1853, N_r/N_b=1.0000)

[5] negativecomm(X1, X2) :- negativebehavior(X1, X2) and negativebehavior(X2, X1) and
    blockpositionindex(X1, X2).
Training Acc: 0.8695 | Eval Acc: 0.8817
Val Coverage: (N_b=111, N_r=100, N_r/N_b=0.9009)
Train Coverage: (N_b=2130, N_r=1860, N_r/N_b=0.8732)

[6] negativecomm(X1, X2) :- negativebehavior(X1, X2) and commonbloc0(X1, X2).
Training Acc: 0.8627 | Eval Acc: 0.8805
Val Coverage: (N_b=98, N_r=93, N_r/N_b=0.9490)
Train Coverage: (N_b=1862, N_r=1671, N_r/N_b=0.8974)

[7] negativecomm(X1, X2) :- negativecomm(X2, X1) and violentactions(X2, X1).
Training Acc: 0.8754 | Eval Acc: 0.8677
Val Coverage: (N_b=127, N_r=102, N_r/N_b=0.8031)
Train Coverage: (N_b=2813, N_r=2250, N_r/N_b=0.7999)

[8] negativecomm(X1, X2) :- negativecomm(X2, X1).
Training Acc: 0.8754 | Eval Acc: 0.8677
Val Coverage: (N_b=127, N_r=102, N_r/N_b=0.8031)
Train Coverage: (N_b=2813, N_r=2250, N_r/N_b=0.7999)

[9] negativecomm(X1, X2) :- negativebehavior(X2, X1) and blockpositionindex(X1, X2) and
    negativecomm(X2, X1).
Training Acc: 0.8595 | Eval Acc: 0.8631
Val Coverage: (N_b=113, N_r=93, N_r/N_b=0.8230)
Train Coverage: (N_b=2111, N_r=1769, N_r/N_b=0.8380)

[10] negativecomm(X1, X2) :- accusation(X1, X2) and accusation(X2, X1).
Training Acc: 0.8757 | Eval Acc: 0.8619
Val Coverage: (N_b=96, N_r=84, N_r/N_b=0.8750)
Train Coverage: (N_b=2060, N_r=1876, N_r/N_b=0.9107)

```

I.3 UMLS DATASET

The *UMLS* dataset consists of biomedical entities connected through a large set of heterogeneous semantic relations. The task requires learning abstract ontological relationships from highly noisy and densely connected knowledge graphs.

$\text{ISA}(X_1, X_2)$

The target predicate $\text{isa}(X_1, X_2)$ captures hierarchical semantic relationships between biomedical concepts. Below, we present representative rules extracted by ANDRE for this predicate. Rules are shown in their raw, prompt-style form as produced by the model.

Listing 6: Extracted Rules for $\text{isa}(X_1, X_2)$ (UMLS Dataset)

```
[1] isa(X1, X2) :- isa(X3, X2) and interacts_with(X1, X3).  
    Training Acc: 0.9449 | Eval Acc: 0.9466  
  
[2] isa(X1, X2) :- isa(X3, X2) and conceptually_related_to(X3, X1).  
    Training Acc: 0.9437 | Eval Acc: 0.9453  
  
[3] isa(X1, X2) :- connected_to(X3, X1) and practices(X3, X2).  
    Training Acc: 0.9437 | Eval Acc: 0.9453  
  
[4] isa(X1, X2) :- not(affects(X3, X1)) and conceptual_part_of(X3, X2).  
    Training Acc: 0.9430 | Eval Acc: 0.9449  
  
[5] isa(X1, X2) :- not(isa(X3, X2)) and conceptual_part_of(X1, X3).  
    Training Acc: 0.9429 | Eval Acc: 0.9445
```

$\text{INTERACTS_WITH}(X_1, X_2)$

The target predicate $\text{interacts_with}(X_1, X_2)$ captures functional, biochemical, or causal interactions between biomedical entities in the UMLS knowledge base. The rules below illustrate how ANDRE infers interaction patterns by leveraging ontological relations (e.g., isa), negated constraints, and higher-order relational dependencies.

Listing 7: Extracted Rules for $\text{interacts_with}(X_1, X_2)$ (UMLS Dataset)

```
[1] interacts_with(X1, X2) :- isa(X2, X1) and not(associated_with(X2, X3)) and not(  
    interacts_with(X2, X1)) and not(ingredient_of(X2, X3)).  
    Training Acc: 0.8920 | Eval Acc: 0.8946  
  
[2] interacts_with(X1, X2) :- isa(X2, X1) and not(interacts_with(X2, X1)) and not(part_of(X3,  
    X1)) and not(measures(X3, X2)).  
    Training Acc: 0.8912 | Eval Acc: 0.8938  
  
[3] interacts_with(X1, X2) :- not(location_of(X3, X2)) and isa(X1, X2) and not(interacts_with(  
    X2, X1)) and not(complicates(X2, X3)).  
    Training Acc: 0.8865 | Eval Acc: 0.8879  
  
[4] interacts_with(X1, X2) :- interacts_with(X1, X3) and interacts_with(X3, X2).  
    Training Acc: 0.8801 | Eval Acc: 0.8823  
  
[5] interacts_with(X1, X2) :- associated_with(X1, X3) and performs(X2, X3).  
    Training Acc: 0.8722 | Eval Acc: 0.8744  
  
[6] interacts_with(X1, X2) :- co_occurs_with(X1, X3) and indicates(X2, X3).  
    Training Acc: 0.8722 | Eval Acc: 0.8744  
  
[7] interacts_with(X1, X2) :- treats(X3, X2) and developmental_form_of(X1, X3).  
    Training Acc: 0.8722 | Eval Acc: 0.8744  
  
[8] interacts_with(X1, X2) :- ingredient_of(X1, X3) and interconnects(X3, X2).  
    Training Acc: 0.8722 | Eval Acc: 0.8744  
  
[9] interacts_with(X1, X2) :- result_of(X2, X3) and adjacent_to(X1, X3).  
    Training Acc: 0.8722 | Eval Acc: 0.8744
```

J RUNTIME COMPARISON

Table 7 reports the total running time required by NTP λ , NeuralLP, DFORL, and ANDRE to generate complete sets of logic programs on the Countries, Nations, and UMLS datasets. The results highlight substantial differences in computational efficiency across methods. NeuralLP consistently achieves the shortest runtime due to its purely neural formulation, while NTP λ incurs significantly higher computational cost, particularly on larger and more relationally complex datasets such as Nations and UMLS. ANDRE exhibits intermediate runtime performance, remaining substantially faster than NTP λ while scaling more favorably than DFORL on dense relational datasets. Overall, these results indicate that ANDRE provides a practical trade-off between computational efficiency and symbolic rule extraction capability.

Table 7: Running time (in minutes) of NTP λ , NeuralLP, DFORL, and ANDRE when generating all logic programs on Countries, Nations, and UMLS datasets. Bold values indicate the best performance.

Dataset	NTP λ	NeuralLP	DFORL	ANDRE
Countries-S1	30.1	0.1	11	3.6
Countries-S2	43.2	0.1	8	3.2
Countries-S3	252.4	0.1	88.4	5.5
Nations	600.4	0.4	0.6	1.2
UMLS	150.4	0.6	5.2	3.5

K FURTHER RESULTS AND BASELINES

Table 8 summarizes the predictive accuracy of CILP++, D-LFIT, DFORL, and ANDRE on the UW-CSE and Alzheimers-amine datasets. On UW-CSE, ANDRE achieves the highest accuracy, substantially outperforming prior symbolic and differentiable rule learning baselines. On Alzheimers-amine, DFORL attains the best performance, while ANDRE achieves competitive accuracy close to the state of the art. These results indicate that ANDRE generalizes effectively across diverse relational domains while maintaining strong predictive performance.

Table 8: Accuracy results (%) on the UW-CSE and Alzheimers-amine datasets. Baseline results are taken from (Gao et al., 2024). Bold values indicate the best performance for each dataset.

Dataset	CILP++	D-LFIT	DFORL	ANDRE
UW-CSE	81.98	79.44	46.90	95.19
Alzheimers-amine	78.70	67.75	99.13	97.04

K.1 ALZHEIMERS-AMINE DATASET

The *Alzheimers-amine* dataset focuses on molecular interaction patterns and chemical property relations relevant to Alzheimer’s disease drug discovery. Predicates encode structural, functional, and physicochemical relationships between molecular components.

`GREAT_NE( $X_1, X_2$ )`

The target predicate `great_ne( $X_1, X_2$ )` models a strong negative interaction between molecular entities. Below, we present representative rules extracted by ANDRE, illustrating both simple logical constraints and more complex recursive and relational dependencies.

Listing 8: Extracted Rules for `great_ne( $X_1, X_2$ )` (Alzheimers-amine Dataset)

```
[1] great_ne(X1, X2) :- not(great_ne(X2, X1)).
    Training Acc: 0.9216 | Eval Acc: 0.9217

[2] great_ne(X1, X2) :- not(great_ne(X2, X1)) and great_ne(X3, X2) and not(r_subst_1(X1, X3)).
    Training Acc: 0.8896 | Eval Acc: 0.8786
```